

GONZALO CRUZ GÓMEZ

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Data Science & Engineering undergraduate (final year, URJC) with a background in Telecommunications Engineering. Builds production-ready data systems: end-to-end ML pipelines, distributed computing, and deep learning. Founder of Asociación Datalab. Seeking an internship in data science or data engineering.

EDUCATION

B.S. Data Science & Engineering URJC

- Final year (expected graduation 2027). Completed two academic years in one (2024–2025). Coursework: ML, Distributed Systems, Deep Learning.

B.S. Telecommunications Engineering UC3M

- 120+ credits in Networking, Physics and Signal Processing. Transferred to specialize in Data Science.

Certificate of Advanced English (C2) Cambridge Assessment

PROJECTS

TripAdvisor Restaurants Data Pipeline

Python, Apache Airflow, PySpark, Kafka, FastAPI, React

- Orchestrated a 5-task Airflow DAG (ETL + ML + streaming) over **1M+ rows** in 50k-row chunks: extract, clean, EDA, preprocess (IncrementalPCA, StandardScaler via `partial_fit`), stream to Kafka.
- Applied PySpark ML KMeans clustering with automatic K selection via silhouette scoring ($K = 10\text{--}80$); OHE or label encoding per column based on cardinality; all parameters in `config.toml`.
- Built dual Spark Structured Streaming consumers: a fast filter path and an ML-scored path computing centroid distances and writing timestamped classifications in real time.
- Developed a full-stack web app (FastAPI + React/Vite): restaurant search via cosine similarity within clusters; live scoring tab pushing KMeans classifications to the browser via Server-Sent Events.

Forest Fire Spread Prediction

R, GLMs, GAMs, Statistical Modeling

- Built GLM and GAM regression models to predict fire-burned area from meteorological covariates; GAMs outperformed GLMs on held-out data with lower residual variance.
- Resolved severe data quality issues (high skew, missing values, near-zero-variance predictors) via log transformation, imputation, and feature selection.
- Applied cross-validation to compare model fits; selected GAMs as the final model based on lower RMSE and better capture of non-linear covariate effects.

Stroke Risk Classification

Python, Scikit-learn, TensorFlow, ANN

- Trained an MLP on an imbalanced medical dataset, applying SMOTE and optimizing the decision threshold to minimize false negatives given the clinical cost of missed strokes.
- Selected AUC-ROC and F1 as evaluation metrics over accuracy; tuned architecture and hyperparameters accordingly.
- Benchmarked the MLP against a logistic regression baseline; the non-linear model improved recall on the stroke class while maintaining acceptable precision.

EXPERIENCE

Founder & President — Asociación Datalab

May 2025 – Present

Established the university's first Data Science student association: defined the strategic roadmap, handled legal registration, and organized technical workshops.

Private Tutor — Mathematics & Physics

Sep 2021 – Present

Mentored high-school students over 5 years, achieving a **100% pass rate** and helping 90% of students improve by at least 3 grade points through personalized plans.

TECHNICAL SKILLS

Programming: Python (Advanced), R, SQL, C

Data Science: Scikit-learn, PyTorch, TensorFlow, Pandas, NumPy, Hadoop, Spark, Matplotlib

Engineering: Apache Airflow, Kafka, Docker, Git/GitHub, Linux (Bash), Claude Code (agentic coding)

Languages: Spanish (Native), English (C2), German (B1), Chinese (HSK3)